

Time Series Analysis of NVIDIA Stock Prices

Time period: 2016 - 2026

Date: May 2, 2026

1. Business Problem

The shares of NVIDIA have been rising steadily but are highly volatile. For an investor practicing dollar-cost averaging, it is important to have an understanding of the trend in the shares and accurate short-term forecasts. This study analyzes the daily closing prices of NVIDIA from January 2016 to December 2025, decomposes price changes, and forecast 30 days ahead.

2. Methodology

The time-series analysis was performed on the basis of NVIDIA's daily closing prices starting from January 4, 2016, and up until December 31, 2025. Data was partitioned into training (excluding 30 last days) and testing (last 30 days of December 2025) sets. The time-series decomposition technique has been used to divide the sequence into trend, seasonality (with 5-day trading week), and residual parts. Next, Exponential Smoothing with an additive trend (Holt's Linear method) has been utilized to make predictions for 60 days, including the test set and next 30 days of January 2026.

3. Results

Figure 1: Time Series Decomposition

The analysis reveals that the prices increase from almost zero in 2016 to above \$200 in 2025. The trend component represents smooth increases. The season component is very small (± 0.05). This means there is no clear weekly pattern. There is random daily variation of $\pm \$10$ in the residual component.

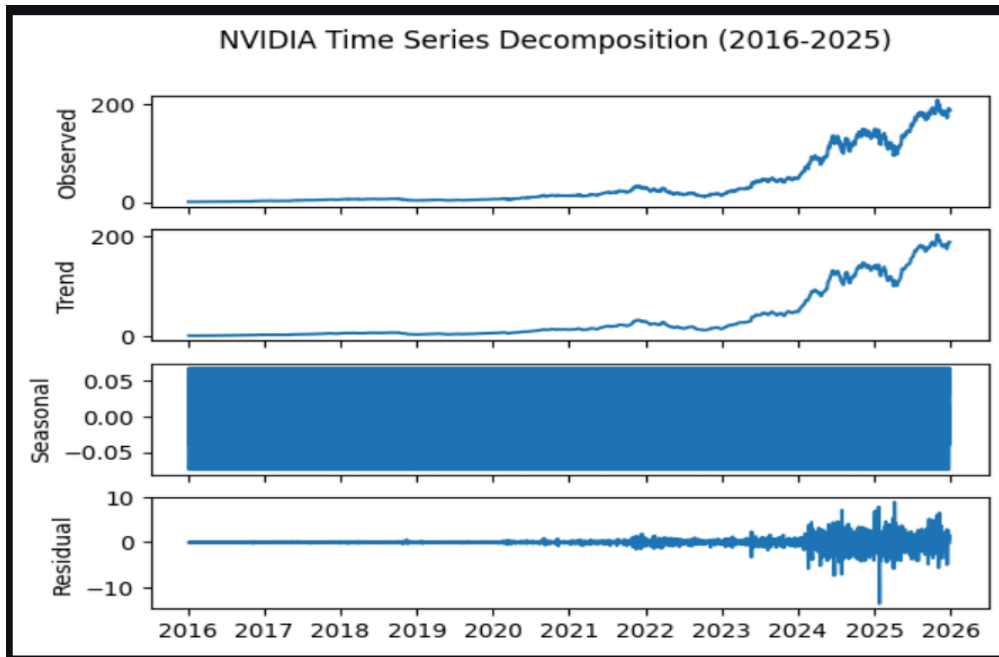
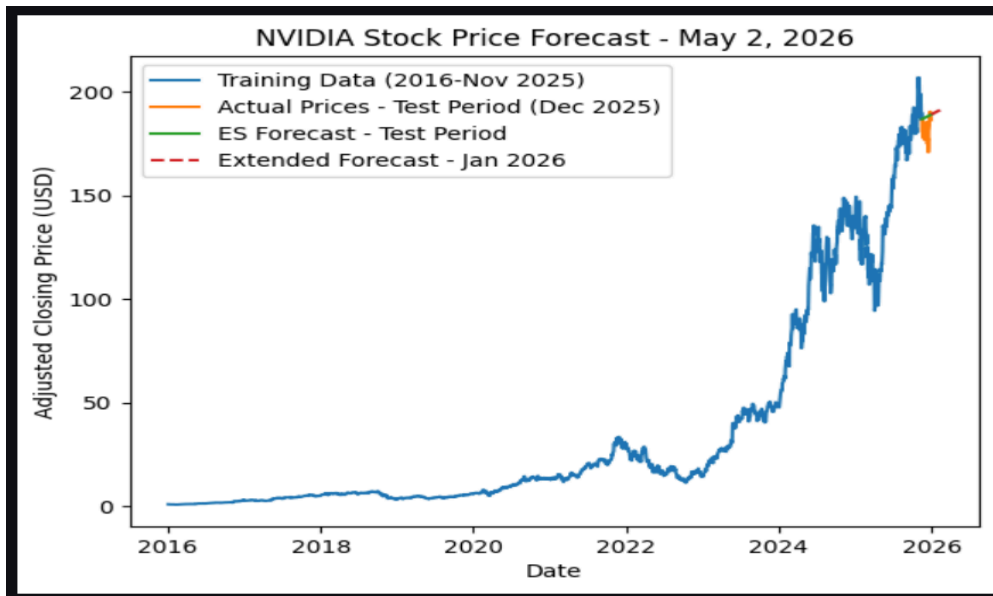


Figure 2: Actual vs Forecast



The prediction graph depicts training data, real prices of December 2025, predicted values for the test set, and projected values for January 2026. The model predicts the trend in prices but fails to predict the noise.

Table 1: Forecast Accuracy Metrics

The model achieved a MAE: 6.2826, RMSE: 7.6304 on the December 2025 test period.

4. Model Justification

The choice of Exponential Smoothing with an additive trend was made due to the presence of a distinct positive trend in the behavior of NVIDIA, which is captured effectively by Holt's model. The prices of the stock do not exhibit any seasonal effects on a week-to-week basis. This forecasting approach focuses on recent prices and is able to cope with volatility without assuming stationarity.

5. Conclusion

Three conclusions can be drawn from the study conducted on dollar cost averaging. Firstly, there is no discernible weekly pattern for NVIDIA stocks, hence making it futile to time purchases based on the day of the week. Secondly, since the stock has been rising strongly, it is evident that purchasing on a consistent basis has been effective in the past and will remain so in the projected period up to January 2026. Thirdly, based on the forecast accuracy measure, the forecast direction can be trusted.